

member id

issue id

member name

student id

publisher name

catagory name

librarian id

issue date

problem: repeating groups exists as member name,student id,employee id same book i

book copy

copy id PK

book id FK

copy status

librarian

librarian id PK

name

person

person id pk

person name

email

phone

address

member

member id pk

person id fk

membership type

student

student id pk

member id fk

employee

employee id pk

member id fk

reservation

reservation id pk

member id fk

book id fk

librarian id fk

reservation date

fine

fine id pk

issue id fk

fine amount

fine status

paid date

person

member

student

employee

person id pk
person name
email
phone
address

member id pk
person id fk
membership type

student id pk
member id fk

employee id pk
member id fk

reservation
reservation id pk
member id fk
book id fk
librarian id fk
reservation date

fine
fine id pk
issue id fk
fine amount
fine status
paid date

publisher
publisher id pk
publisher name

catagory
catagory id pk
catagory name

library unnormal form

employee id	book id	book title	copy id
due date	return date	fine amount	reservation date
d,name ,catagory repeats and author fine and reservation info also repeats			

first normal from

issue	reservation	member	book
issue id PK	reservation id PK	member id PK	book id pk
member id FK	member id FK	member name	title
copy id FK	book id FK	student id FK	author name
librarian id FK	reservation date	employee id FK	publisher name
issue date			catagory name
due date			
return date			
fine amount			

second normal form

librarian	book	author	book author
librarian id pk	book id pk	author name	book id pk
person id fk	book title	author id pk	author id fk
	publisher name		
	catagory name		

third normal form

librarian	book	author	book author
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librarian id PK
person id fk

book id PK
book title
publisher name
catagory name

author name
author id PK

book id pk
author id fk

copy status

author name

book copy

copy id pk
book id fk
copy status

issue

issue id pk
member id fk
copy id fk
librarian id fk
issue date
due date
return date

book copy

issue

copy id pk
book id fk
copy status

issue id pk
member id fk
copy id fk
librarian id fk
issue date
due date
return date